

Multi PCSM

IEC

Easy maintenance.

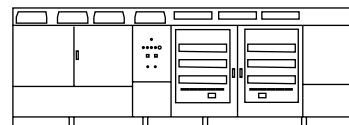
Integrated MV solution in the same enclosure.

Up to 4 independent DC inputs.

Advanced grid support.

Compatible with all battery technologies.

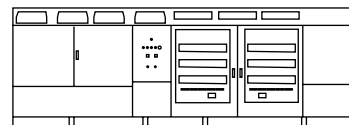




REFERENCES	FP4999MH2	FP4999MH4	FP5020MH2	FP5020MH4	
AC	AC Output Power (kVA/kW) @30 °C ^[1]	4999	5240		
	AC Output Power (kVA/kW) @35 °C ^[1]	4999	5020		
	AC Output Power (kVA/kW) @40 °C ^[1]	4800			
	AC Output Power (kVA/kW) @50 °C ^[1]	4360			
	Operating Grid Voltage (kV)	33 kV ±10%	34.5 kV ±10%		
	Rated Impulse Withstand Voltage	150 kV	145 kV		
	Operating Grid Frequency (Hz)	50 Hz	60 Hz		
	Current Harmonic Distortion (THDi)	<3% per IEEE 519			
	Power Factor (CosPhi) ^[2]	0.5 leading ... 0.5 lagging			
	Reactive Power Compensation	Four quadrant operation			
	Overload Capability	166% - 100 ms / 150% - 5 s / 120% - 8 s / 110% - 15 s			
	DC	DC Voltage Range ^[3]	976 V – 1500 V		
Maximum DC Voltage		1500 V			
DC Voltage Ripple		<3%			
Max. DC Continuous Current (A)		2740	1370	2740	1370
Max. DC Short Circuit Current per Input (kA)		500 kA with a time constant of 1 ms			
Battery Technology		All type of batteries (BMS required)			
Number of Separated DC Inputs		2	4	2	4
EFFICIENCY & AUX. SUPPLY	Efficiency (Max) (η)	98.00% including MV transformer			
	CEC (η)	97.53% including MV transformer			
CABINET	Dimensions [WxDxH] (ft)	21.3 x 6.5 x 7.5			
	Dimensions [WxDxH] (m)	6.5 x 2.0 x 2.3			
	Weight (lbs)	30865			
	Weight (kg)	14000			
	Type of Ventilation	Forced air cooling			
ENVIRONMENT	Degree of Protection	IP55			
	Operating Temperature Range ^[4]	From -40 °C to +60 °C, >30 °C power derating			
	Operating Relative Humidity Range	From 4% to 100%			
	Storage Temperature Range	From -40 °C to +60 °C			
	Max. Altitude (above sea level) ^[5]	2000 m			
CONTROL	Communication Protocol	Modbus TCP			
	Grid Forming (VISMA) / Black Start Capable	Standard			
PROTECTIONS	Ground Fault Protection	Insulation monitoring device			
	Humidity Control	Active heating			
	General AC Protection & Disconnection	36 kV MV switchgear (2L+V)			
	General DC Protection & Disconnection	High-speed fuses, Motorized DC disconnect switches ^[6]			
	Overvoltage Protection	Type II for AC and Type I+II for DC			
STANDARDS	Safety	IEC 62109-1 / IEC 62109-2 / IEC 62477-1 / IEC 62477-2			
	Utility Interconnect ^[7]	IEC 62116 / G99 / VDE 4110-4120-4130 / CEI 0-16 / NTS 2.1 / EN 50549			

NOTES

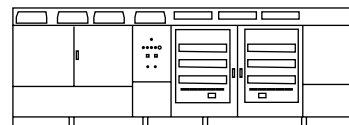
- [1] Values at 1.00-Vac nom and CosPhi=1. Consult Power Electronics for derating curves.
[2] Consult P-Q charts available: $Q(kVar) = \sqrt{(S(kVA))^2 - P(kW)^2}$.
[3] Consult Power Electronics for derating curves. In the event of overvoltage in the grid, the minimum DC voltage will vary proportionally with the AC voltage.
[4] Optional available for temperatures below -25 °C.
[5] Consult Power Electronics for altitudes above 1000 m.
[6] Battery short circuit disconnection must be done on the battery side.
[7] Consult Power Electronics for other applicable standards / grid codes.



REFERENCES	FP5021MH2	FP5021MH4	FP5022MH2	FP5022MH4	
AC	AC Output Power (kVA/kW) @30 °C ^[1]	5240			
	AC Output Power (kVA/kW) @35 °C ^[1]	5020			
	AC Output Power (kVA/kW) @40 °C ^[1]	4800			
	AC Output Power (kVA/kW) @50 °C ^[1]	4360			
	Operating Grid Voltage (kV)	33 kV ±10%	30 kV ±10%		
	Rated Impulse Withstand Voltage	150 kV	145 kV		
	Operating Grid Frequency (Hz)	50 Hz			
	Current Harmonic Distortion (THDi)	<3% per IEEE 519			
	Power Factor (CosPhi) ^[2]	0.5 leading ... 0.5 lagging			
	Reactive Power Compensation	Four quadrant operation			
	Overload Capability	166% - 100 ms / 150% - 5 s / 120% - 8 s / 110% - 15 s			
DC	DC Voltage Range ^[3]	976 V – 1500 V			
	Maximum DC Voltage	1500 V			
	DC Voltage Ripple	<3%			
	Max. DC Continuous Current (A)	2740	1370	2740	1370
	Max. DC Short Circuit Current per Input (kA)	500 kA with a time constant of 1 ms			
	Battery Technology	All type of batteries (BMS required)			
	Number of Separated DC Inputs	2	4	2	4
EFFICIENCY & AUX. SUPPLY	Efficiency (Max) (η)	98.00% including MV transformer			
	CEC (η)	97.53% including MV transformer			
CABINET	Dimensions [WxDxH] (ft)	21.3 x 6.5 x 7.5			
	Dimensions [WxDxH] (m)	6.5 x 2.0 x 2.3			
	Weight (lbs)	30865			
	Weight (kg)	14000			
	Type of Ventilation	Forced air cooling			
	ENVIRONMENT	Degree of Protection	IP55		
Operating Temperature Range ^[4]		From -40 °C to +60 °C, >30 °C power derating			
Operating Relative Humidity Range		From 4% to 100%			
Storage Temperature Range		From -40 °C to +60 °C			
Max. Altitude (above sea level) ^[5]		2000 m			
CONTROL	Communication Protocol	Modbus TCP			
	Grid Forming (VISMA) / Black Start Capable	Standard			
PROTECTIONS	Ground Fault Protection	Insulation monitoring device			
	Humidity Control	Active heating			
	General AC Protection & Disconnection	36 kV MV switchgear (2L+V)			
	General DC Protection & Disconnection	High-speed fuses, Motorized DC disconnect switches ^[6]			
	Overvoltage Protection	Type II for AC and Type I+II for DC			
STANDARDS	Safety	IEC 62109-1 / IEC 62109-2 / IEC 62477-1 / IEC 62477-2			
	Utility Interconnect ^[7]	IEC 62116 / G99 / VDE 4110-4120-4130 / CEI 0-16 / NTS 2.1 / EN 50549			

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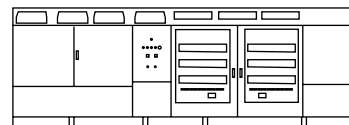
- [1] Values at 1.00-Vac nom and CosPhi=1. Consult Power Electronics for derating curves.
[2] Consult P-Q charts available: $Q(kVar) = \sqrt{(S(kVA))^2 - P(kW)^2}$.
[3] Consult Power Electronics for derating curves. In the event of overvoltage in the grid, the minimum DC voltage will vary proportionally with the AC voltage.
[4] Optional available for temperatures below -25 °C.
[5] Consult Power Electronics for altitudes above 1000 m.
[6] Battery short circuit disconnection must be done on the battery side.
[7] Consult Power Electronics for other applicable standards / grid codes.



REFERENCES		FP5023MH	FP5023MH2	FP5023MH4
AC	AC Output Power (kVA/kW) @30 °C ^[1]	5240		5360
	AC Output Power (kVA/kW) @35 °C ^[1]	5020		5150
	AC Output Power (kVA/kW) @40 °C ^[1]	4800		4940
	AC Output Power (kVA/kW) @50 °C ^[1]	4360		4520
	Operating Grid Voltage (kV)	20 kV ±10%		
	Rated Impulse Withstand Voltage	125 kV		
	Operating Grid Frequency (Hz)	50 Hz		
	Current Harmonic Distortion (THDi)	<3% per IEEE 519		
	Power Factor (CosPhi) ^[2]	0.5 leading ... 0.5 lagging		
	Reactive Power Compensation	Four quadrant operation		
	Overload Capability	166% - 100 ms / 150% - 5 s / 120% - 8 s / 110% - 15 s		
DC	DC Voltage Range ^[3]	976 V - 1500 V		
	Maximum DC Voltage	1500 V		
	DC Voltage Ripple	<3%		
	Max. DC Continuous Current (A)	5478	2740	1370
	Max. DC Short Circuit Current per Input (kA)	500 kA with a time constant of 1 ms		
	Battery Technology	All type of batteries (BMS required)		
EFFICIENCY & AUX. SUPPLY	Number of Separated DC Inputs	1	2	4
	Efficiency (Max) (η)	98.00% including MV transformer		
	CEC (η)	97.53% including MV transformer		
CABINET	Dimensions [WxDxH] (ft)	21.3 x 6.5 x 7.5		
	Dimensions [WxDxH] (m)	6.5 x 2.0 x 2.3		
	Weight (lbs)	30865		
	Weight (kg)	14000		
	Type of Ventilation	Forced air cooling		
ENVIRONMENT	Degree of Protection	IP55		
	Operating Temperature Range ^[4]	From -40 °C to +60 °C, >30 °C power derating		
	Operating Relative Humidity Range	From 4% to 100%		
	Storage Temperature Range	From -40 °C to +60 °C		
	Max. Altitude (above sea level) ^[5]	2000 m		
CONTROL	Communication Protocol	Modbus TCP		
	Grid Forming (VISMA) / Black Start Capable	Standard		
PROTECTIONS	Ground Fault Protection	Insulation monitoring device		
	Humidity Control	Active heating		
	General AC Protection & Disconnection	24 kV SF6 Free MV switchgear (2L+V)		
	General DC Protection & Disconnection	High-speed fuses, Motorized DC disconnect switches ^[6]		
	Overvoltage Protection	Type II for AC and Type I+II for DC		
STANDARDS	Safety	IEC 62109-1 / IEC 62109-2 / IEC 62477-1 / IEC 62477-2		
	Utility Interconnect ^[7]	IEC 62116 / G99 / VDE 4110-4120-4130 / CEI 0-16 / NTS 2.1 / EN 50549		

NOTES

- [1] Values at 1.00-Vac nom and CosPhi=1. Consult Power Electronics for derating curves.
[2] Consult P-Q charts available: $Q(kVar) = \sqrt{(S(kVA))^2 - P(kW)^2}$.
[3] Consult Power Electronics for derating curves. In the event of overvoltage in the grid, the minimum DC voltage will vary proportionally with the AC voltage.
[4] Optional available for temperatures below -25 °C.
[5] Consult Power Electronics for altitudes above 1000 m.
[6] Battery short circuit disconnection must be done on the battery side.
[7] Consult Power Electronics for other applicable standards / grid codes.



REFERENCES		FP5026MH2	FP5026MH4
AC	AC Output Power (kVA/kW) @30 °C ^[1]	5240	
	AC Output Power (kVA/kW) @35 °C ^[1]	5020	
	AC Output Power (kVA/kW) @40 °C ^[1]	4800	
	AC Output Power (kVA/kW) @50 °C ^[1]	4360	
	Operating Grid Voltage (kV)	33 kV ±10%	
	Rated Impulse Withstand Voltage	150 kV	
	Operating Grid Frequency (Hz)	60 Hz	
	Current Harmonic Distortion (THDi)	<3% per IEEE 519	
	Power Factor (CosPhi) ^[2]	0.5 leading ... 0.5 lagging	
	Reactive Power Compensation	Four quadrant operation	
	Overload Capability	166% - 100 ms / 150% - 5 s / 120% - 8 s / 110% - 15 s	
DC	DC Voltage Range ^[3]	976 V - 1500 V	
	Maximum DC Voltage	1500 V	
	DC Voltage Ripple	<3%	
	Max. DC Continuous Current (A)	2740	1370
	Max. DC Short Circuit Current per Input (kA)	500 kA with a time constant of 1 ms	
	Battery Technology	All type of batteries (BMS required)	
EFFICIENCY & AUX. SUPPLY	Efficiency (Max) (η)	98.00% including MV transformer	
	CEC (η)	97.53% including MV transformer	
CABINET	Dimensions [WxDxH] (ft)	21.3 x 6.5 x 7.5	
	Dimensions [WxDxH] (m)	6.5 x 2.0 x 2.3	
	Weight (lbs)	30865	
	Weight (kg)	14000	
	Type of Ventilation	Forced air cooling	
ENVIRONMENT	Degree of Protection	IP55	
	Operating Temperature Range ^[4]	From -40 °C to +60 °C, >30 °C power derating	
	Operating Relative Humidity Range	From 4% to 100%	
	Storage Temperature Range	From -40 °C to +60 °C	
	Max. Altitude (above sea level) ^[5]	2000 m	
CONTROL	Communication Protocol	Modbus TCP	
	Grid Forming (VISMA) / Black Start Capable	Standard	
PROTECTIONS	Ground Fault Protection	Insulation monitoring device	
	Humidity Control	Active heating	
	General AC Protection & Disconnection	36 kV MV switchgear (2L+V)	
	General DC Protection & Disconnection	High-speed fuses, Motorized DC disconnect switches ^[6]	
	Overvoltage Protection	Type II for AC and Type I+II for DC	
STANDARDS	Safety	IEC 62109-1 / IEC 62109-2 / IEC 62477-1 / IEC 62477-2	
	Utility Interconnect ^[6]	IEC 62116 / G99 / VDE 4110-4120-4130 / CEI 0-16 / NTS 2.1 / EN 50549	

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